

Task-identity structure and interpopulation connectivity shape compositional geometry in recurrent neural networks

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Abstract

Compositional representations provide a candidate mechanism for reusing learned task components across contexts. We investigate how task-identity encoding and interpopulation connectivity shape this organization in recurrent neural networks trained on four tasks crossing sensory modality (color or shape) with response axis (1 or 2). We compare 100 networks with structured task identities and 100 with orthogonal identities, and examine connectivity dependence in networks partitioned into four populations. We characterize task-conditioned activity using principal component analysis, alignment of task-difference vectors, and the residual of an additive prediction of one task centroid from the other three. We formalize this residual as the modality-by-axis interaction in a balanced two-factor representation, distinguishing directional alignment from exact additivity. Structured task identities yield higher alignment and lower compositional prediction error than orthogonal identities. In the connectivity sweep, orthogonal networks exhibit greater alignment at the fully connected endpoint than at the disconnected endpoint, whereas structured networks remain comparatively stable. These results suggest that explicit task-factor information and recurrent communication are associated with approximately additive task geometry. Our findings concern representational compositionality in trained tasks; whether this geometry supports generalization to untrained task combinations remains an open question.

Keywords: recurrent neural networks; compositionality; task representations; principal component analysis; structural modularity

1 Introduction

Flexible behavior requires the ability to reuse knowledge across tasks that share some, but not all, of their computational demands. One candidate organizing principle is compositionality: a task can be represented or implemented by combining components that also participate in other tasks. This principle motivates a distinction between *representational compositionality*, in which task-related neural states have a reusable structure, and *behavioral compositional generalization*, in which a system successfully performs a previously untrained combination. The former is a possible substrate for the latter, but is not by itself a behavioral demonstration.

Multitask recurrent neural networks (RNNs) provide a framework for investigating this distinction. Yang and colleagues showed that networks trained on multiple cognitive tasks can develop related task representations and can, in some settings, use algebraic combinations of task instructions [1]. Riveland and Pouget demonstrated that natural-language instructions can organize sensorimotor representations in ways that support performance on unseen tasks [2]. These results motivate examining whether explicitly structured, nonlinguistic task cues likewise favor shared representational geometry.

A complementary question concerns network connectivity. Constraining recurrent connections provides a means of manipulating the routes through which information is exchanged between neural populations. Frameworks for training biologically constrained RNNs make such manipulations possible [3]. However, structural separation does not necessarily imply functional specialization: B  na and Goodman showed that specialization depends on environmental structure, resource constraints, and communication between modules [4]. These findings caution against treating anatomical or architectural modularity as equivalent to compositional computation.

The present task family is inspired by work examining shared sensory and motor subspaces in monkeys performing compositionally related tasks [5]. We consider the full crossing of two sensory modalities and two response axes, yielding four tasks. We manipulate whether task identity is supplied as an orthogonal cue or a structured cue organized into named input fields. We also vary interpopulation connectivity in four-population networks under both encoding conditions to examine how recurrent communication influences the resulting representations.

Our central question is whether these manipulations are associated with an approximately additive organization of task-conditioned activity. We express that question using a common mathematical framework: task-difference alignment tests whether a factor induces similar directions across contexts, whereas an additive prediction residual tests whether those differences agree in both direction and magnitude. Together, these analyses connect the geometry of task representations to the information supplied by task cues and the connectivity available for recurrent processing.

2 Methods

2.1 Experimental design

We trained 200 networks for the task-identity comparison: 100 with structured task identities and 100 with orthogonal task identities. Trials comprised an intertrial interval (ITI), a fixation period (FIX), and a stimulus period (STIM), occupying the time-index ranges 0–10, 10–15, and 16–40, respectively. We recorded activity over 40 time points. STIM onset and the single-time-point prediction-error comparison were at $t_0 = 16$; this analysis time was not a shared PCA reference for the other time points.

To examine connectivity dependence, we trained networks partitioned into four neural populations at five nominal interconnectivity levels,

$$C = \{0, 0.25, 0.50, 0.75, 1\}. \quad (1)$$

We trained 30 networks per task-identity condition at each connectivity level, giving $2 \times 5 \times 30 = 300$ networks across the sweep.

2.2 Factorial task structure and task-identity encodings

Let $m \in \{C, S\}$ denote color or shape and $a \in \{1, 2\}$ denote response axis. The task set is

$$\mathcal{T} = \{C1, C2, S1, S2\} = \{C, S\} \times \{1, 2\}. \quad (2)$$

Here $C1$ and $C2$ denote color-guided responses along saccade axes 1 and 2, respectively; $S1$ and $S2$ denote shape-guided responses along the same two axes. Both stimulus attributes varied continuously. The parameter θ_m specifies the modulation of attribute m , represented numerically by

$$s_m = \cos(\theta_m), \quad m \in \{C, S\}. \quad (3)$$

The task identity specified the relevant feature and response axis. The feature modulation s_m is distinct from the task-identity cue defined below.

Each network receives sensory input together with a cue identifying the task. In the orthogonal condition, distinct task encodings are mutually orthogonal. For nonzero task vectors $\mathbf{q}_{ma}^{\text{orth}} \in \mathbb{R}^{d_q}$, this means

$$(\mathbf{q}_{ma}^{\text{orth}})^T \mathbf{q}_{m'a'}^{\text{orth}} = 0 \quad \text{whenever } (m, a) \neq (m', a'). \quad (4)$$

One-hot vectors are a possible realization, but orthogonality does not require one-hot coordinates or unit-length vectors. In the structured condition, task information is organized into named input fields. The field names identify the meaning of the inputs, while the numerical field values specify the cue presented to the network. This organization does not by itself imply that the cue vectors have equal norms or satisfy an additive identity.

2.3 Recurrent networks and interpopulation connectivity

For network r , let $\mathbf{h}_{r,k,t} \in \mathbb{R}^{N_r}$ denote its hidden activity on trial k at time t , where N_r is the number of recurrent neurons. We express the recurrent computation as

$$\mathbf{h}_{r,k,t+1} = F_{\theta_r}(\mathbf{h}_{r,k,t}, \mathbf{x}_{k,t}, \mathbf{q}_{m_k a_k}; W_r^{\text{rec}}), \quad \hat{\mathbf{y}}_{r,k,t} = G_{\theta_r}(\mathbf{h}_{r,k,t}). \quad (5)$$

Here $\mathbf{x}_{k,t}$ is the sensory input, θ_r denotes the trainable parameters, and G_{θ_r} is the readout.

For the connectivity manipulation, each network comprised four populations of five neurons, giving $N_r = 20$. Assign neuron i to population $g(i) \in \{1, 2, 3, 4\}$. Each population was internally fully connected. Interpopulation masks were randomly generated using Gaussian sampling and held fixed throughout training. The resulting retained-connection pattern can be represented by

$$W_r^{\text{rec}}(c) = M_r(c) \odot \tilde{W}_r^{\text{rec}}, \quad (6)$$

where $M_r(c)$ is a binary mask and $\odot$ denotes elementwise multiplication. The interpopulation density of a realized mask is

$$\hat{c}_r = \frac{\sum_{i,j: g(i) \neq g(j)} [M_r(c)]_{ij}}{\#\{(i, j) : g(i) \neq g(j)\}}. \quad (7)$$

We define $c = 0$ as no interpopulation connections and $c = 1$ as full interpopulation connectivity, distinguishing the nominal parameter c from the realized density $\hat{c}_r$. Intrapopulation connectivity remained fully connected across the sweep. Increasing c therefore reduces structural separation rather than increasing structural modularity. Fixing the mask constrains which connections can be trained; it does not imply that the retained recurrent weights are fixed.

2.4 Task-conditioned activity and principal component analysis

The activity analyzed by PCA is organized by neuronal channel, time point, and trial. Let P_r be the number of recorded neuronal channels and L_r the number of trials for network r . With 40 recorded time points, the activity tensor is

$$\mathcal{H}_r \in \mathbb{R}^{P_r \times 40 \times L_r}, \quad (8)$$

with entries $[\mathcal{H}_r]_{i,t,k}$ giving the activity of recorded channel i at time t on trial k . The eight-channel recordings have $P_r = 8$. We distinguish the recorded-channel count P_r from the total network size N_r ; they are equal when the full recurrent state is recorded. Let $\mathbf{a}_{r,k,t} \in \mathbb{R}^{P_r}$ denote the recorded activity vector. PCA reduces its neuronal dimension to three while preserving the time and trial indexing. All cosine similarities, VPDs, and centroid-prediction MSEs were calculated using these three retained components.

Let $\mathcal{K}_{r,ma}$ be the trials for task ma in network r , with $n_{r,ma} = |\mathcal{K}_{r,ma}|$ and $L_r = \sum_{ma \in \mathcal{T}} n_{r,ma}$. The recorded-activity task centroid is

$$\bar{\mathbf{a}}_{r,ma}(t) = \frac{1}{n_{r,ma}} \sum_{k \in \mathcal{K}_{r,ma}} \mathbf{a}_{r,k,t}. \quad (9)$$

Differences in stimulus or response-category sampling can contribute to apparent task geometry, making the trial composition used to estimate these centroids important for interpretation.

We fit PCA separately for each network and each individual time point, and transform the activity at that time point using its own fitted basis. At time t , form $X_r(t) \in \mathbb{R}^{L_r \times P_r}$ with one trial activity vector per row, pooling all four tasks. Let $\bar{\mathbf{x}}_r(t)$ be the pooled mean at that time and define

$$X_{r,c}(t) = X_r(t) - \mathbf{1}\bar{\mathbf{x}}_r(t)^\top, \quad (10)$$

$$\Sigma_r(t) = \frac{1}{L_r - 1} X_{r,c}(t)^\top X_{r,c}(t), \quad (11)$$

$$\Sigma_r(t) \mathbf{u}_{r,d}(t) = \lambda_{r,d}(t) \mathbf{u}_{r,d}(t), \quad (12)$$

with orthonormal eigenvectors and ordered eigenvalues $\lambda_{r,1}(t) \geq \dots \geq \lambda_{r,P_r}(t) \geq 0$. The eigenvectors are principal directions; the eigenvalues quantify variance along those directions. For D retained dimensions, write $U_{r,D}(t) = [\mathbf{u}_{r,1}(t), \dots, \mathbf{u}_{r,D}(t)]$ and

$$\begin{aligned} \mathbf{z}_{r,k}(t) &= U_{r,D}(t)^\top (\mathbf{a}_{r,k,t} - \bar{\mathbf{x}}_r(t)), \\ \mu_{r,ma}(t) &= U_{r,D}(t)^\top (\bar{\mathbf{a}}_{r,ma}(t) - \bar{\mathbf{x}}_r(t)). \end{aligned} \quad (13)$$

The explained-variance fraction is

$$\text{EVR}_{r,D}(t) = \frac{\sum_{d=1}^D \lambda_{r,d}(t)}{\sum_{d=1}^{P_r} \lambda_{r,d}(t)}, \quad (14)$$

provided the total variance is positive. We retain $D = 3$ components at each time point, so $\Sigma_r(t) \in \mathbb{R}^{P_r \times P_r}$, $U_{r,3}(t) \in \mathbb{R}^{P_r \times 3}$, and each projected activity vector lies in $\mathbb{R}^3$.

All task contrasts within a network and time point use that time point's shared PCA basis. Because the mean, loadings, and retained subspace can change with t , the resulting projections are separate time-specific representations rather than a trajectory in one fixed coordinate system. We summarize task geometry over time through scalar metrics calculated within each time point, rather than interpreting movement between independently fitted PC coordinates. Temporal differences in projected metrics can reflect changes in the retained subspace as well as changes in neural activity.

2.5 Factor contrasts and directional alignment

For clarity, suppress the network and time indices and let $\mu_{ma} \in \mathbb{R}^D$ denote a task centroid. We name each contrast by the factor that *changes*:

$$\mathbf{d}_1^{\text{mod}} = \mu_{S1} - \mu_{C1}, \quad \mathbf{d}_2^{\text{mod}} = \mu_{S2} - \mu_{C2}, \quad (15)$$

$$\mathbf{d}_C^{\text{axis}} = \mu_{C2} - \mu_{C1}, \quad \mathbf{d}_S^{\text{axis}} = \mu_{S2} - \mu_{S1}. \quad (16)$$

For nonzero vectors $\mathbf{v}$ and $\mathbf{w}$, define

$$\cos(\mathbf{v}, \mathbf{w}) = \frac{\mathbf{v}^\top \mathbf{w}}{\|\mathbf{v}\|_2 \|\mathbf{w}\|_2} \in [-1, 1]. \quad (17)$$

The two alignment scores are

$$A_{\text{mod}} = \cos(\mathbf{d}_1^{\text{mod}}, \mathbf{d}_2^{\text{mod}}), \quad A_{\text{axis}} = \cos(\mathbf{d}_C^{\text{axis}}, \mathbf{d}_S^{\text{axis}}). \quad (18)$$

A value of 1 denotes alignment, 0 orthogonality, and -1 opposition. A zero-length contrast makes the corresponding score undefined, not zero. High alignment indicates a shared direction but does not establish matching contrast magnitudes or orthogonality between the two factors.

In Fig. 2, panel headings identify the factor held fixed: “Modality” compares within-modality, axis-changing contrasts, whereas “Axis” compares within-axis, modality-changing contrasts. Equation (16) instead names the factor that changes. Reversing both vectors in a cosine comparison leaves the score unchanged.

2.6 Additive task geometry and vector-predicted difference

An additive representation has the form

$$\mu_{ma} = \mathbf{b} + \mathbf{u}_m + \mathbf{v}_a, \quad (19)$$

where $\mathbf{b}$ is a shared baseline and the two factors contribute independently. Such a representation predicts the fourth centroid as

$$\widehat{\mu}_{S2} = \mu_{S1} - \mu_{C1} + \mu_{C2}. \quad (20)$$

Define the interaction residual and vector-predicted difference (VPD) by

$$\boldsymbol{\delta} = \mu_{S2} - \widehat{\mu}_{S2} = \mu_{S2} - \mu_{S1} - \mu_{C2} + \mu_{C1}, \quad \text{VPD} = \|\boldsymbol{\delta}\|_2. \quad (21)$$

Small VPD indicates a small departure from an additive parallelogram, in the coordinate system and activity scale being analyzed. VPD is not a measure of behavioral task error.

Proposition 1 (Equivalent descriptions of additive task geometry). *For four task centroids in a common vector space, the following are equivalent: (i) an additive decomposition (19) exists; (ii) $\boldsymbol{\delta} = \mathbf{0}$; (iii) the two modality contrasts are equal; and (iv) the two response-axis contrasts are equal. With equal weight on the four centroids, the minimum summed squared residual of an additive fit is $\|\boldsymbol{\delta}\|_2^2 / 4$.*

The proof is given in Appendix A. In particular,

$$\boldsymbol{\delta} = \mathbf{d}_2^{\text{mod}} - \mathbf{d}_1^{\text{mod}} = \mathbf{d}_S^{\text{axis}} - \mathbf{d}_C^{\text{axis}}. \quad (22)$$

Consequently, VPD combines directional and magnitude discrepancies, whereas cosine similarity captures only direction. Additivity does not require the modality and response-axis effects to be perpendicular. Conversely, a collapsed representation in which all four centroids coincide has zero VPD but carries no centroid-level task information. Task separability and behavioral performance are therefore essential complementary checks.

PCA truncation can conceal nonadditivity. If $\boldsymbol{\delta}_a$ is the residual in the unprojected recorded-activity space, then

$$\boldsymbol{\delta}_z = U_D^T \boldsymbol{\delta}_a, \quad \|\boldsymbol{\delta}_a\|_2^2 = \|\boldsymbol{\delta}_z\|_2^2 + \|(I - U_D U_D^T) \boldsymbol{\delta}_a\|_2^2. \quad (23)$$

Thus a small three-dimensional VPD need not imply a small residual in the unprojected activity. When all neurons are recorded, this is also the full hidden-state residual. The same issue can alter cosine similarities. Full-state and retained-dimension sensitivity analyses are needed to establish that the findings are not specific to the visualization subspace.

2.7 Prediction-error scaling and the scrambled control

The unnormalized coordinate-wise mean squared prediction error is

$$E_{\text{raw}} = \frac{1}{D} \sum_{d=1}^D (\mu_{S2,d} - \widehat{\mu}_{S2,d})^2 = \frac{\text{VPD}^2}{D}. \quad (24)$$

Let s_d^2 denote a pooled score variance in coordinate d , estimated from a specified reference sample. A dimensionless coordinate-standardized error is

$$E_{\text{std}} = \frac{1}{D} \sum_{d=1}^D \frac{(\mu_{S2,d} - \widehat{\mu}_{S2,d})^2}{s_d^2}, \quad s_d^2 > 0, \quad (25)$$

which divides residuals by standard deviations before squaring. Dividing residuals by variances before squaring instead weights squared errors by inverse squared variance and is not invariant to uniform rescaling of activity. Near-zero variances require an exclusion or regularization rule.

A complementary rotation- and scale-invariant centroid statistic follows from Proposition 1:

$$\eta_{\text{int}} = \frac{\|\delta\|_2^2}{4 \sum_{ma \in \mathcal{T}} \|\mu_{ma} - \bar{\mu}\|_2^2}, \quad \bar{\mu} = \frac{1}{4} \sum_{ma \in \mathcal{T}} \mu_{ma}. \quad (26)$$

For nonzero total centroid variation, $0 \leq \eta_{\text{int}} \leq 1$; it is the fraction of that variation unexplained by the equal-weight additive fit. It is undefined when all centroids coincide. We introduce this statistic as a theoretical characterization of interaction strength; the empirical comparisons below use cosine similarity, VPD, and MSE rather than estimates of η_{int} .

We compare the additive prediction with the scrambled predictor

$$\widehat{\mu}_{S2}^{\text{scr}} = \mu_{S1} + \mu_{C1} - \mu_{C2}. \quad (27)$$

This is a deterministic reversed-contrast control, not a sampled null distribution. For an exactly additive configuration with $\mu_{C1} = \mathbf{b}$, $\mu_{S1} = \mathbf{b} + \mathbf{u}$, $\mu_{C2} = \mathbf{b} + \mathbf{v}$, and $\mu_{S2} = \mathbf{b} + \mathbf{u} + \mathbf{v}$, its residual is

$$\mu_{S2} - \widehat{\mu}_{S2}^{\text{scr}} = 2\mathbf{v}. \quad (28)$$

The control therefore introduces an error proportional to a task-factor effect even under exact additivity. Better performance than this control supports the chosen algebraic relation, but does not alone establish significance against arbitrary noncompositional geometries. A permutation-based extension would require a null hypothesis and an exchangeability scheme that preserve relevant stimulus and response structure.

2.8 Aggregation and statistical analysis

We summarize time-resolved alignment and VPD across 100 networks per task-identity condition using means and standard deviations. These bands describe between-network variation rather than confidence intervals for the mean. The trained network is the unit of analysis; repeated time points and individual neurons are not treated as independent network replicates. Additive and scrambled prediction errors are paired within each network.

We used paired Wilcoxon signed-rank tests to compare additive prediction errors between structured and orthogonal conditions and to compare additive and scrambled errors within each condition. This nonparametric procedure evaluates paired differences using the ranks of their absolute magnitudes [6]. For the within-condition comparisons, each pair consists of two predictions for the same network. Numerical results are given in Table 1.

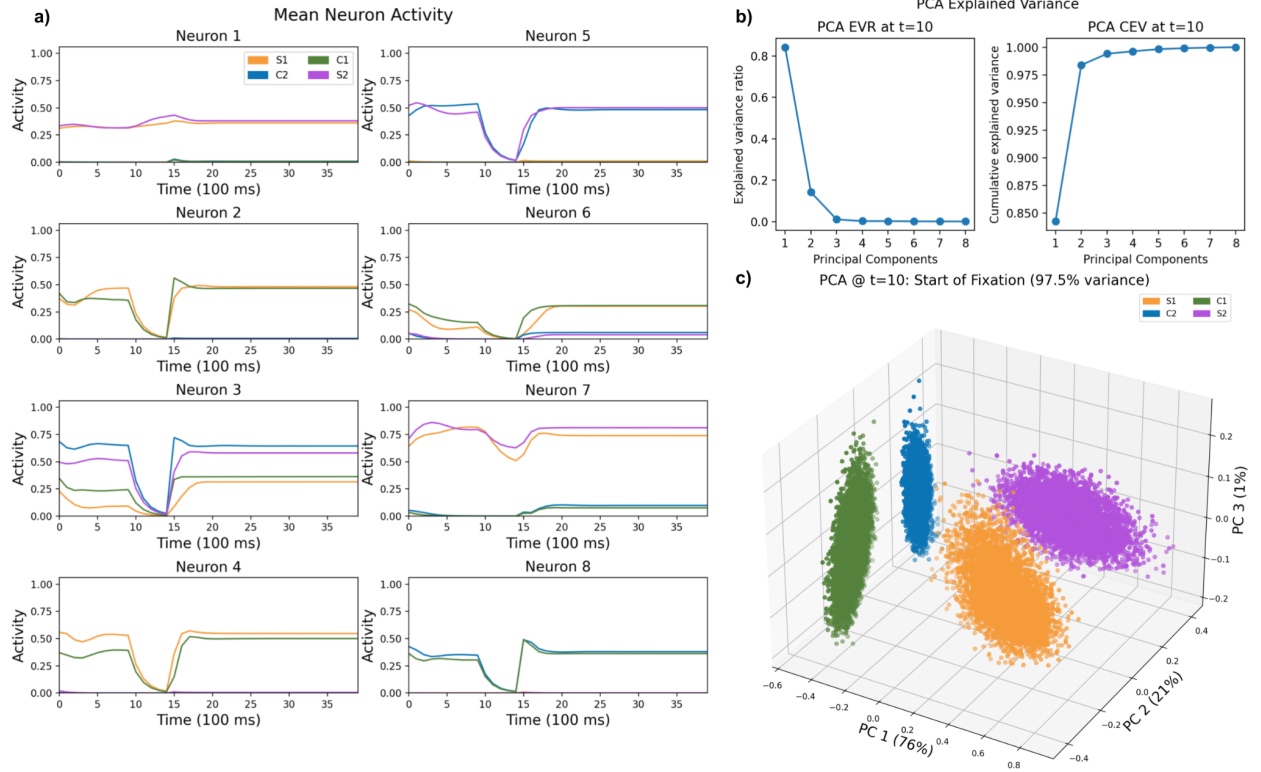


Figure 1: Activity and PCA in a representative structured network. (a) Task-conditioned mean activity for eight recorded units. (b) Individual and cumulative explained-variance curves. (c) Eight-dimensional neuronal activity projected into three principal components at $t = 10$. Colors distinguish the four modality–response-axis combinations.

3 Results

3.1 Low-dimensional views reveal task-conditioned activity

The representative structured network in Fig. 1 exhibits task-dependent activity in individual units and distinguishable clusters in a three-dimensional PCA projection. These displays motivate a centroid-based description of task geometry. They do not establish that every network has the same organization or that three components retain all interaction-related information. The illustrated snapshot uses PCA fitted at $t = 10$ and is distinct from the prediction-error comparison at stimulus onset, $t_0 = 16$.

3.2 Structured task identities are associated with more additive geometry

Across the time-resolved summaries, structured networks show greater alignment than orthogonal networks in both cosine comparisons (Fig. 2a). The difference is visible over much of the trial, including the relatively stable period after the stimulus-onset reference. Both encoding conditions retain some alignment, so the contrast is one of degree rather than a categorical presence or absence of compositional structure.

Structured networks also show lower mean VPD than orthogonal networks (Fig. 2b). In conjunction with the alignment results, this supports a closer approximation to the additive centroid relation. However, unnormalized VPD is sensitive to activity scale, and a three-dimensional residual can miss discrepancies in discarded dimensions. The joint trend is therefore descriptive evidence for more additive geometry in the analyzed representation, not proof of a scale-independent or full-state effect.

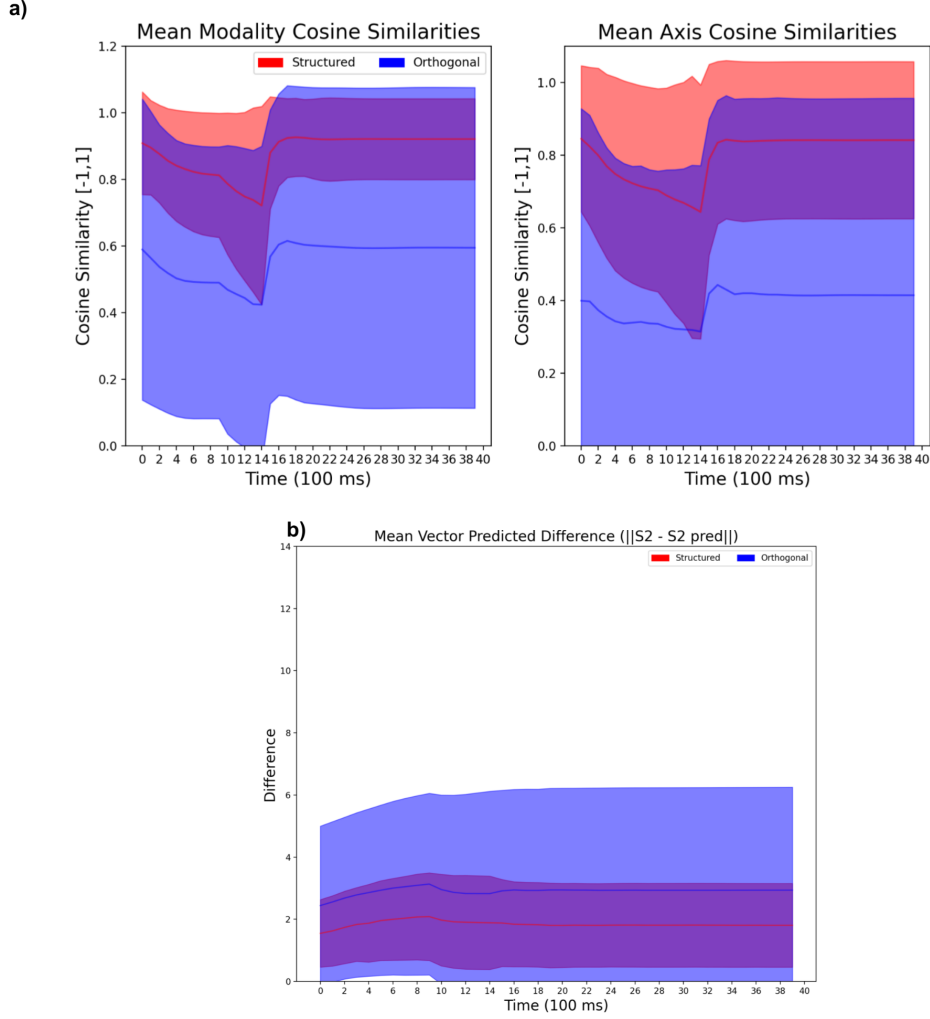


Figure 2: Time-resolved task geometry under structured and orthogonal encoding. (a) Cosine similarities, with structured networks in red and orthogonal networks in blue. The “Modality” panel compares within-modality, response-axis contrasts; the “Axis” panel compares within-axis, modality contrasts, as defined in Eq. (16). (b) VPD for the additive prediction of $S2$. Curves and shaded bands show the mean and standard deviation across 100 networks per encoding condition.

Table 1: Paired Wilcoxon signed-rank comparisons of centroid-prediction errors at $t_0 = 16$. Each encoding condition comprises 100 networks.

Comparison	p -value
Orthogonal additive vs. structured additive	6.21×10^{-9}
Structured additive vs. structured scrambled	2.00×10^{-14}
Orthogonal additive vs. orthogonal scrambled	1.22×10^{-9}

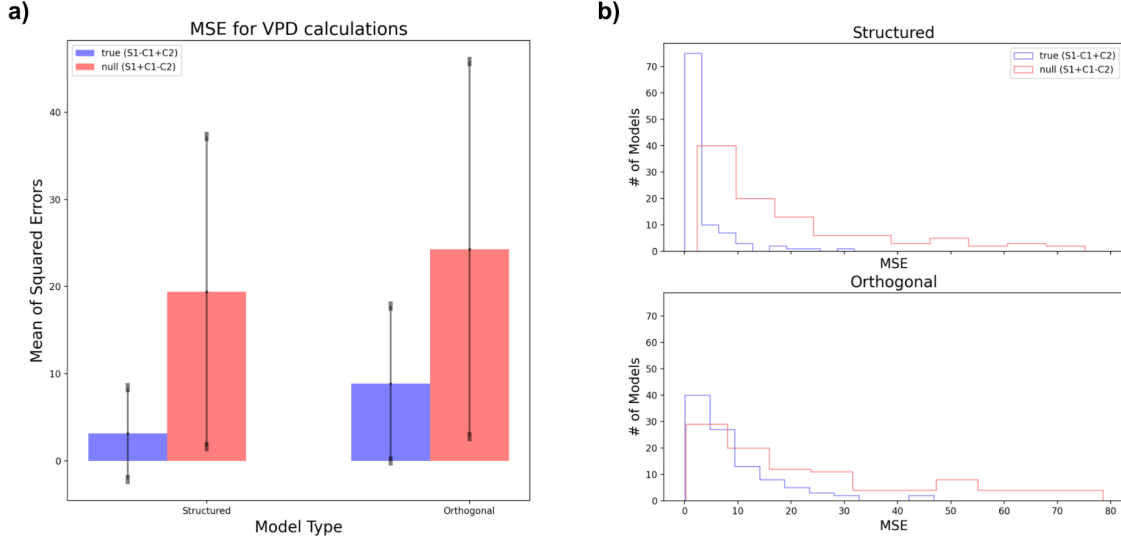


Figure 3: Additive and scrambled prediction errors. (a) Mean MSE for the additive and scrambled predictors under each encoding condition at $t_0 = 16$. (b) Error distributions across 100 networks per condition. Blue (“true”) denotes the additive predictor, and red (“null”) denotes the deterministic scrambled control; these colors indicate prediction type rather than task-identity condition. Paired Wilcoxon signed-rank results appear in Table 1.

At stimulus onset ($t_0 = 16$), structured networks have lower mean additive prediction error than orthogonal networks, and additive errors are lower than scrambled-control errors in both encoding conditions (Fig. 3). The histograms show substantial asymmetry and heterogeneity among networks. Together with the time-resolved alignment and VPD results, these comparisons indicate that structured cues are associated with task centroids that more closely approximate the additive prediction.

Paired Wilcoxon signed-rank comparisons yielded $p = 6.21 \times 10^{-9}$ for orthogonal versus structured additive errors, $p = 2.00 \times 10^{-14}$ for additive versus scrambled errors in structured networks, and $p = 1.22 \times 10^{-9}$ for additive versus scrambled errors in orthogonal networks (Table 1). These tests concern the prediction errors in Fig. 3; they do not constitute a test of the encoding-by-connectivity interaction.

3.3 Connectivity dependence differs between encoding conditions

We next examined how task-identity encoding interacts with connectivity in networks partitioned into four populations. Structured networks have lower MSE than orthogonal networks at all five nominal connectivity levels (Fig. 4). The orthogonal curve is lower at $c = 1$ than at $c = 0$, but is not monotonic over the intermediate levels.

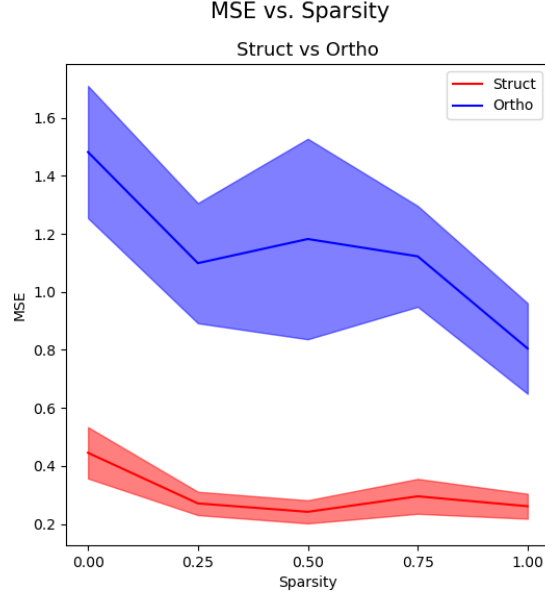


Figure 4: Error across the four-population connectivity sweep. MSE across five connectivity levels for structured (red) and orthogonal (blue) task-identity conditions, with 30 networks per condition at each level. The horizontal axis, labeled “Sparsity,” denotes interpopulation connectivity c : 0 indicates no interpopulation connections and 1 indicates full interpopulation connectivity. *The error and shaded-band definitions await confirmation (Q5–Q6).*

The cosine comparisons provide a direct assessment of task geometry across connectivity levels (Fig. 5). Structured networks maintain comparatively high alignment across the sweep. Orthogonal networks have higher alignment at $c = 1$ than at $c = 0$ in both panels, although the right-hand curve shows a marked intermediate decrease. Thus the data support an endpoint improvement, not a strictly monotonic dependence on interconnectivity. Orthogonal networks also remain below the structured curves at the fully connected endpoint.

Taken together, the sweep is consistent with a larger dependence of orthogonal-cue geometry on interpopulation communication, while structured-cue geometry is comparatively stable. A direct encoding-by-connectivity interaction analysis would quantify this difference. Under the definition of c in Methods, the endpoint improvement accompanies *less* structural separation rather than greater modularity.

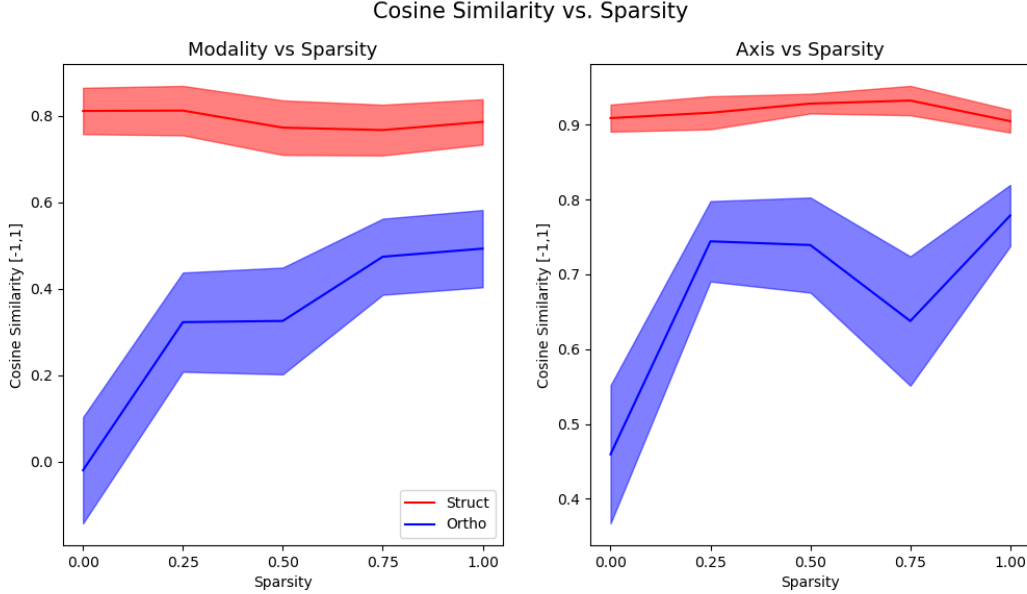


Figure 5: Task-vector alignment across the connectivity sweep. Cosine similarities for the two task-factor comparisons in four-population networks, with 30 networks per encoding condition at each connectivity level. Structured networks are shown in red and orthogonal networks in blue. The horizontal axes, labeled “Sparsity,” denote interpopulation connectivity c , increasing from disconnected to fully connected populations. *The mapping of panel headings to the contrasts in Eq. (16) and the shaded-band definition await confirmation (Q4, Q6).*

4 Discussion

4.1 Task cues and recurrent communication as distinct constraints

Our results suggest that cue structure and recurrent connectivity are associated with different aspects of task representation. Structured cues organize task information through named fields, whereas orthogonal cues provide mutually perpendicular task-identity vectors. If the field values encode shared task factors additively, a linear input mapping can preserve that relation. Additive hidden-state geometry could then partly reflect structure already present in the cues rather than the discovery of a compositional computation. Named fields alone do not establish this algebraic property; it depends on the numerical encoding. This is a mechanistic hypothesis, not a result established by the present analyses.

Greater interpopulation connectivity may allow orthogonal-cue networks to coordinate task factors across populations. However, the sweep changes opportunities for communication and may also change the number of trainable parameters or effective recurrent gain. Without parameter-matched and performance-matched controls, these alternatives cannot be separated. The observed relative stability of structured networks is consistent with reduced dependence on communication for expressing their task geometry, but should not be interpreted as proof that communication is unnecessary.

These distinctions align with the broader separation between structural modularity, functional specialization, and compositional task organization [4]. A network can reuse task factors without assigning each factor to a distinct anatomical population. Conversely, sparsely interconnected populations need not implement reusable computations. The present centroid metrics characterize representational organization, not module-specific causal roles.

4.2 What additive geometry establishes

The interaction residual gives a precise interpretation of the task-vector arithmetic: zero residual is equivalent to an additive representation of the four task centroids. The residual also quantifies the least-squares cost of an additive fit. This formalization makes clear why cosine similarity and VPD are complementary rather than interchangeable.

Nevertheless, centroid additivity is a restricted notion of compositionality. It does not characterize within-task distributions, nonlinear transformations, or the temporal sequence of computations. The shared sensory and motor subspaces examined in the motivating neural work [5] concern information represented across tasks, not simply the arrangement of four task means. Moreover, a network can exhibit additive centroids while implementing distinct task-specific computations elsewhere in its state space. The absence of held-out-task testing precludes a direct claim of compositional generalization.

4.3 Limitations and decisive follow-up analyses

Several controls would strengthen the mechanistic interpretation. First, comparisons in the full hidden-state space and across multiple retained dimensions would establish whether the geometric effects extend beyond the visualization subspace. Second, examining task separability and behavioral performance alongside interaction residuals would help distinguish useful factorization from a reduction in task-related information. Sensitivity to activity scaling and variance normalization is also important when comparing prediction errors between networks.

Third, trained networks should be compared with untrained networks, cue-only responses, and a linear input-projection baseline to distinguish inherited cue geometry from training-dependent organization. Fourth, a leave-one-combination-out design should train on three tasks and evaluate the fourth, rotating the held-out task and specifying how its instruction is supplied. Such a design would directly address the gap between the present representation-level observations and behavioral generalization [1, 2]. For example, with one-hot encoding, an unseen cue activates input weights that received no task-specific training, so comparisons must account for differences in the information supplied by the cue.

Finally, comparisons with matched task performance, parameter counts, and recurrent scaling would help isolate the effects of interpopulation communication. Population-specific perturbations could then test whether the putatively shared factors have a causal role in task execution. These steps would determine whether the current associations reflect reusable computations, an input-geometry effect, or differences in capacity and optimization.

5 Conclusion

Structured task identities are associated with greater task-vector alignment and smaller additive prediction errors than orthogonal task identities. Our connectivity manipulation further suggests that orthogonal-cue geometry is more sensitive to interpopulation communication, with better endpoint alignment under greater connectivity and a persistent gap from structured networks. Formalizing task-vector arithmetic as a two-factor interaction connects these observations to a precise notion of additive representation. Together, our findings identify task-cue structure and recurrent connectivity as candidate influences on representational compositionality, motivating direct tests of whether this organization supports generalization to new task combinations.

Data and code availability

All project data and code, including trained models, recorded activity, and numerical analysis outputs, remain private and are not publicly available.

A Additive decomposition and interaction residual

Proof of Proposition 1. An additive decomposition gives $\delta = \mathbf{0}$ by cancellation. Conversely, if $\delta = \mathbf{0}$, set $\mathbf{b} = \mu_{C1}$, $\mathbf{u}_C = \mathbf{v}_1 = \mathbf{0}$, $\mathbf{u}_S = \mu_{S1} - \mu_{C1}$, and $\mathbf{v}_2 = \mu_{C2} - \mu_{C1}$. These choices reproduce all four centroids. The equivalences with equal contrasts follow directly from Eq. (16).

For the least-squares statement, use effect coding $\sigma_C = -1$, $\sigma_S = 1$, $\rho_1 = -1$, and $\rho_2 = 1$. Every four-centroid configuration has a unique expansion

$$\mu_{ma} = \mathbf{b} + \sigma_m \mathbf{u} + \rho_a \mathbf{v} + \sigma_m \rho_a \mathbf{w}, \quad (29)$$

where

$$\mathbf{b} = \frac{1}{4}(\mu_{C1} + \mu_{C2} + \mu_{S1} + \mu_{S2}), \quad (30)$$

$$\mathbf{u} = \frac{1}{4}(-\mu_{C1} - \mu_{C2} + \mu_{S1} + \mu_{S2}), \quad (31)$$

$$\mathbf{v} = \frac{1}{4}(-\mu_{C1} + \mu_{C2} - \mu_{S1} + \mu_{S2}), \quad (32)$$

$$\mathbf{w} = \frac{1}{4}(\mu_{C1} - \mu_{C2} - \mu_{S1} + \mu_{S2}) = \frac{1}{4}\delta. \quad (33)$$

The four scalar coding columns 1, σ , ρ , and $\sigma\rho$ are orthogonal over the balanced task set. The equal-weight least-squares additive fit therefore removes only the interaction term. Its residual at each centroid is $\sigma_m \rho_a \mathbf{w}$, and the summed squared residual is

$$\text{SSE}_{\text{add}} = 4 \|\mathbf{w}\|_2^2 = \frac{1}{4} \|\delta\|_2^2. \quad (34)$$

Similarly, the total centered sum of squares is

$$\text{SST} = 4(\|\mathbf{u}\|_2^2 + \|\mathbf{v}\|_2^2 + \|\mathbf{w}\|_2^2). \quad (35)$$

Thus Eq. (26) equals $\text{SSE}_{\text{add}}/\text{SST}$ and lies in $[0, 1]$ whenever $\text{SST} > 0$. $\square$

This derivation uses equal weights on the four task centroids, regardless of their trial counts. An uncertainty-weighted fit would require a separately specified objective and would not in general have the same residual formula. For any common linear projection P , the projected interaction is $P\delta$. Equation (23) follows by the orthogonal decomposition into the span of U_D and its complement.

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